# FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS
UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE



GUSTAV SIMONSEN
TEACHING ASSISTANT

# WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46.

Every year, projects end up built entirely on the screen and the speaker.

Not because touch was the wrong choice. Because nobody knew that a coin motor costs 12 kr and takes two wires.

# HOW TODAY WORKS

**Six benches. Already wired. Already running.**

- Ten minutes each, on a hard timer.
- You change one parameter and answer one question.
- You wire nothing. That is what the project weeks are for.

**Then you build one signal and someone else tries to read it.**

# RUN SHEET

Times are offsets from the start. The rotation is the spine of it, so keep it moving.

| | | |
|---|---|---|
| **0:00** | 10 min | Two motors, same message. Which felt urgent? |
| **0:10** | 60 min | Bench rotation, 10 minutes each |
| **1:10** | 10 min | Round-the-room: what surprised you |
| **1:20** | 30 min | Build one signal |
| **1:50** | 10 min | Blind test and pack down |

# THE SIX BENCHES

# 01 ERM COIN MOTOR

CUTANEOUS · VIBRATION

An off-centre weight on a motor shaft. The same part that is in your phone and every game controller.

Turn the knob and notice what you cannot do. Speed and strength are mechanically welded together, so "gentle but fast" is not available to you.

| | |
|---|---|
| part | 10mm 3V coin ERM |
| drive | 2N2222 + 1N4148 |
| pin | D9 (PWM) |
| draw | ~75 mA |
| spin-up | 20–40 ms |
| cost | ~12 kr |

CARD QUESTION

**At what point does it stop feeling like information and start feeling like a malfunction?**

# 02 LRA + PIEZO DISC

CUTANEOUS · VIBRATION

Two ways out of the ERM's compromise.

The LRA hits resonance in about 5 ms and stops just as fast, which is why phone keyboards use it.

The piezo is a crystal that flexes when you put current across it. Near-instant, almost no power, but shallow. A tick, not a thump.

```
parts    LRA 10mm + piezo
driver   DRV2605L (I2C)
pin   A4 SDA / A5 SCL
effects   123 waveforms
cost   ~90 kr
```

CARD QUESTION

**Which of the three vibrators would you actually put on a wrist, and why not the others?**

# 03 SOLENOID TAP

CUTANEOUS · IMPACT

A push-pull solenoid firing a single 15 ms pulse against a fingertip.

This is the bench everyone remembers. A single knock carries urgency that no amount of buzzing does, because it reads as a person tapping you rather than a machine signalling.

```
part    5V push-pull
drive    MOSFET + flyback
pin    D6
peak    ~1.1 A
duty    <= 25%, gets hot
cost    ~45 kr
```

CARD QUESTION

**How many taps before it goes from alert to nagging?**

# 04 CAPACITIVE TOUCH

KINESTHETIC · TOUCH-TO-ACTUATE

Touch a surface with a bare finger and the servos react.

There is no button and no sensor you can point at. The input is a wire taped behind a plate, read on an analogue pin.

Any surface can become an input this way: foil, card, fabric.

```
sense     ADCTouch on A0
baseline  25-sample roll
trigger   1.01x average
debounce  100 ms
servos    D4 + D13
cost      ~0 kr, a wire
```

CARD QUESTION

**Put your hand near the plate without touching. When exactly did it decide that was a touch?**

# BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire. The thinking is where it costs you.

**reading > baselineAverage() * 1.01**

**The threshold is RELATIVE, never absolute.**

• Readings drift with humidity, with mains hum, with how you are sitting. A fixed threshold works in the morning and fails after lunch.

**The baseline only learns while untouched.**

• Otherwise a long press slowly teaches it that a finger is normal, and the touch quietly stops registering.

# 05 PELTIER WARM / COOL

CUTANEOUS · THERMAL

A thermoelectric tile: heats one side, cools the other, reverses when you flip the current.

Sit with it. It takes several seconds before you are sure which way it went.

And alternating hot with cold does not read as a pattern. It just reads as lukewarm.

```
part   TEC1-12706
drive   L298N H-bridge
pin   D3 PWM / D4-5
draw   2–4 A, bench PSU
onset   3–8 s
cap   45 C in software
```

CARD QUESTION

**Time yourself. How long until you would bet money on warmer vs cooler?**

# 06 THE SAME SIGNAL, TWICE

INTRAMODAL · AUDIO + HAPTIC

One transducer on a plate, playing 40 Hz you feel and 400 Hz you hear, both from the same driver.

Toggle between them alone and together.

Together is not louder. It is more certain. Sending the same thing down two channels is the cheapest reliability your project can buy.

```
part    bone-conduction
amp    PAM8403 class-D
pin    D11 (tone)
felt    ~40 Hz
heard    ~400 Hz
cost    ~110 kr
```

CARD QUESTION

**With ear defenders on, does the signal still work?**

# INPUTS, AND PRIOR ART

# INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire on an analogue pin. See bench 04.
- Heart rate (PPG). Steady at rest, useless once the hand moves.
- Skin conductance (GSR). Drifts constantly, so relative change only.
- Flex sensor. The cheap route to a data glove.
- Pressure (FSR). How hard, not just whether. Calibrate each pad.
- Your own phone. Accelerometer, gyro, mic, camera, vibration motor. No soldering at all.

# TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

**Capacitive sensing, servos and actuators (Arduino Uno)**

• A moving screen that breathes when idle and retreats when you touch it, with a browser control panel over the serial port.

• Take away: easing and idle motion are what separate "a servo moved" from "it reacted to me". Both are a dozen lines.

**Instrumented sock (ESP32, bachelor project)**

• Six force sensors under a foot, sent to a server over Wi-Fi in batches, each with its own calibration constants.

• Take away: calibrate per sensor, and batch your writes.

# BUILD ONE SIGNAL

# THE BRIEF

30 minutes. Deliberately not long enough to be precious about it.

Pick one actuator from the rotation.

Encode three messages: arrived, something wrong, finished.

Vary only two things: the RHYTHM of the pulses, and how STRONG they are. No swapping actuators between messages.

Hand it to another group with the screen turned away. They name all three.

Write down which two got confused, and what would have separated them.

# THE CONSTRAINT THAT MATTERS

**The recipient cannot look at the device.**

If your design only works when someone is watching a screen, you built a visual interface with a motor glued to it.

# GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop